Emergency Training Simulator POC

Markdown documentation PDF copy

Environment Object Contract

Source: docs/architecture/environment-object-contract.md

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Boundary: this project is still a POC, not operational guidance.

Environment Object Contract

This contract keeps environment objects shaped the same way across the backend, logs, reviews, and future clients.

The backend still owns scenario state. These objects are not real 3D assets and do not run real physics.

Required shared fields

Every environment object should include:

- object_id: the key used in environment.objects
- object_type: what kind of object it is, like wooden_chair or door
- label: short human-readable name
- location: simple scenario location label
- blocks_path: whether it currently blocks movement

Optional shared metadata

Use these when helpful:

- collision_enabled: future-facing collision note
- collision_shape: simple future collision shape label
- mass_kg_estimate: rough future mass estimate
- physics_enabled: whether the scenario claims active physics; currently false
- physics_note: plain-language note about what is and is not simulated
- force_preview: future-facing notes about possible push/pull directions

These fields are metadata only right now. They do not create real gravity, collision, mass, or force behavior.

Current object-specific fields

Wooden chair:

- moved: whether the chair has been moved aside

Entry door:

- open_state: closed or open
- locked: metadata only for now
- door_type: metadata only for now
- hinge_side: metadata only for now
- swing_direction: metadata only for now

Current scenario objects

- wooden_chair_poc: starts near the entry path and blocks movement until move_obstacle.
- entry_door_poc: starts closed and blocks room entry until open_door.

Blender review rule

When generated assets are introduced later, no asset should enter assets/reviewed/ or Unreal until it passes a manual Blender review checklist.